AI based Face Recognition for Smart Attendance System

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***Abstract*—Process of manual attendance in colleges and schools is time consuming and maintaining the record is burden to teachers. Attendance details may be manipulated at any time by the teachers or students. In the manual attendance system, there is a possibility of making proxy attendance by students and making wrong entries by the teacher. This paper proposes "Smart attendance system using image processing techniques" as a replacement for conventional system of attendance. The model is developed with the help of real-time OpenCV library. Most represented a part of the body that can differentiate one person among the others is face. The smart attendance system compares the query images with the previous database images consists of student's images which have been saved at the time of enrollment. At the predefined slots as per the timetable, 4 to 5 images are captured by high resolution digital camera/webcam and these images are compared with the database. With the help of Image processing techniques and artificial intelligence, the attendance report being generated. The complete information is stored in excel format and also sent to the respective teacher's mail. The proposed system is experimented with conditions such as different lighting, head positions, distance between the student and cameras. Hence, the proposed system is a suitable replacement for manual attendance system. The system needs less equipment and is cost-efficient.**

Keywords—* *attendance monitoring, Smart attendance system, face recognition, AI based image processing

I. Introduction

Attendances of students are being maintained by every school, college and university for a group of students in each class. Empirical evidences have shown that there is a significant correlation between students' attendances and their academic performances. There is also a claim stated that students who have poor attendance records will generally link to poor academic performances. Therefore, faculty has to maintain proper record for the attendance. Manual attendance record system is in-efficient and requires more time to record and maintain the attendance of every student. Analyzing the attendance requires manual calculations, which may also time consuming and may have errors.

Hence there is a requirement of a system that will provide a better solution for student attendance in educational institutes. In many Institutes, the process of taking attendance is done by the faculty members by asking them to respond according to their roll numbers which is time consuming process. So, adoption of methods like biometrics such as finger print scanning, face recognition will help to save time. But for a class with a greater number of students, fingerprint scanning is again time consuming and cannot be a suitable solution. Another way of identifying the student's presence is by face recognition. The face image data of the students are compared with photography image taken during class hours and presence of student is identified. This is achieved by the collaboration of the artificial intelligence techniques and image processing techniques.

The main objective is to develop face recognition based automated student attendance system. In order to achieve better performance, the test images and training images of this proposed approach are limited to frontal and upright facial images that consist of a single face only. The test images and training images have to be captured by using the same device to ensure no quality difference. The students have to register in the database with their identity such as Roll Number and Name. The enrolment can be done through the user-friendly interface.

The smart attendance system using digital image processing that automates the attendance tracking process. The benefits such as accuracy improvement, administrative workload reduction, ensuring reliable and efficient management of attendance records in educational institutions and simplify the transfer of attendance record to stack holders. As the data is available online, the analysis of each student, overall class etc. can be done easily.

II. Methods of Attendance Tracking

A. Manual Attendance System

A manual attendance system involves recording attendance by physically checking who is present and documenting this information through methods like paper registers, roll calls, sign-in sheets, or punch cards. While effective in smaller settings, these systems can be time-consuming, error-prone, and less efficient compared to digital systems, making them harder to manage, store, and analyse in larger organizations. Students or employees can fake attendance by having someone else respond on their behalf. This practice reduces the reliability of manual attendance.

B. Biometric based Systems

A biometric attendance system uses unique biological traits like fingerprints, facial recognition, or iris scans to verify an individual's identity for recording attendance. It offers real-time data, easy tracking, and management, making it highly efficient for both small and large organizations. Even though, it has the drawback that include the high cost of implementation, potential privacy concerns among users, the need for maintenance and updates, and possible technical issues that can arise, such as failures in recognizing individuals due to changes in physical appearance or environmental conditions.

This system also has drawbacks such as a) Hygiene Concerns – Physical contact with finger print scanners raises health and hygiene issues, especially in the post-pandemic era. b) Recognition Failures – Dirt, sweat, or cuts on fingers can reduce scanning accuracy. c) Time Delays – In high-traffic environments, fingerprint-based systems cause long queues and delays [1, 2].

C. RFID Card based System

An RFID card-based attendance system uses Radio Frequency Identification (RFID) technology to automatically record attendance. Each individual is issued an RFID card containing a unique identifier. When the card is brought near an RFID reader, the reader captures the card's information and records the individual's presence. While RFID card-based attendance systems are efficient and reliable, they have some drawbacks. These include the possibility of card loss or theft, leading to unauthorized access. The initial setup and maintenance costs can be high, especially for larger organizations. Users might also face issues if the RFID reader fails to detect the card due to technical glitches or physical barriers [1, 2].

III. Materials And Methods

The proposed system integrates hardware and software to automate attendance tracking using python code and hardware prototype implemented using Raspberry Pi 4 and Pi camera module. The system captures group photos and processes them with image processing libraries such as OpenCV and face recognition. It employs machine learning algorithms to detect and recognize student faces, comparing them against a pre-stored dataset.

Recognized students are marked as present in an Excel sheet, updated automatically, while unidentified faces are marked as absent. Additionally, the system features an email reporting function to send attendance reports ensuring accurate and timely documentation and communication of attendance records. By combining hardware and advanced image processing, the system offers an efficient solution for automated attendance tracking using digital image processing [1, 3].

The developed system has the following elements.

Hardware Integration

- *RaspberryPi4:* Using a RaspberryPi4 as the central processing unit for the system. The Pi will handle image processing, student identification, and communication with other components.
- *Pi Camera Module:* Integration of a PiCamera Module to capture group photos. The camera will be used to take pictures of students during attendance sessions.

Smart Photo Processing

- *Development of Image Processing Libraries:* Utilization of image processing libraries such as OpenCV and face recognition to process group photos. These libraries will handle tasks like face detection, recognition, and preprocessing of images to ensure accurate identification.

Data Analytics Algorithms

- *Integration of Machine Learning Algorithms:* Implementation of machine learning algorithms to recognize and identify students' faces from the group photos. These algorithms will analyze facial features and match them with a pre- stored dataset of student images to determine identities.

Face Detection and Recognition

- *Detection Algorithms:* Deployment of face detection algorithms to locate faces within the group photos. These algorithms will identify the presence and position of faces in the images.
- *Recognition Algorithms:* Utilization of face recognition algorithms to match detected faces with the dataset of student images. This will involve feature extraction and comparison techniques to accurately identify each student.

Data Storage and Retrieval

- *Excel Sheet Integration:* Integration with Excel sheets using libraries such as pandas and openpyxl to store and retrieve student data. The system will open the Excel sheet, update attendance records, and save changes automatically.

Attendance Marking

- *Automated Attendance Updates:* The system will automatically mark students as present or absent based on the recognition results. Detected and recognized faces will be marked as present, while unidentified faces will be marked as absent.

Email Reporting

- *Automated Email Notifications:* Implementation of an email reporting feature using the smtplib library to send attendance reports. The system will compose an email with the attendance details and send it to the user automatically.

The process has the following steps (Figure 1):

1. *Initialize the Device and Camera*

The system starts by initializing the required hardware and software components. This step includes:

- Powering on the device and ensuring that all necessary drivers and system libraries are loaded.
- Activating the camera module and testing its functionality.
- Verifying network connectivity (if required) to ensure proper image processing and database access.
- Checking system logs and previous session data to prevent potential errors.

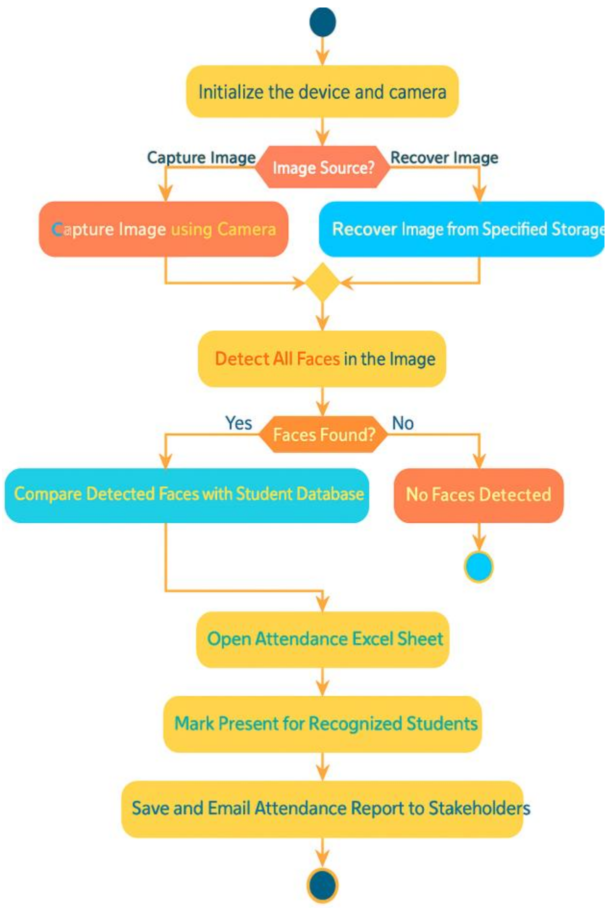


Fig. 1. Process Flow of Attendance System

If any issues arise, the system will generate error messages and prompt the administrator to resolve them before proceeding.

2. *Image Source Selection*

This step determines whether an image needs to be captured or if an existing mage should be used. The two options include:

Capture Image using Camera: The system accesses the live camera feed to take a snapshot of the students present in the classroom. The captured image is stored temporarily for processing.

Recover Image from Specified Storage: If a pre-captured image is available (such as from a previous session), the system retrieves it from a specified storage location. This may include a local database, cloud storage, or an external device.

Proper image selection is crucial as it ensures that the system processes the correct attendance data.

3. *Face Detection*

Once an image is available, the system proceeds to detect face using an advanced AI- based algorithm. The process involves:

Pre-processing the Image: This step enhances image quality through contrast adjustment, noise reduction, and gray scale conversion to improve face detection accuracy.

Face Localization: The algorithm scans the image to locate faces using bounding boxes. It identifies key facial features such as eyes, nose, and mouth.

Multi-Face Handling: If multiple faces are detected, the system separates each one for individual processing.

Handling Occlusions: The system can filter out obstructions like masks, glasses, or poor lighting conditions to ensure accurate face detection.

If no faces are detected, the system halts further operations and prompts for an image input.

4. *Face Recognition and Comparison*

Once faces are detected, they are compared against a pre-registered student database. This involves:

Feature Extraction: The system extracts unique facial features such as distances between eyes, jaw structure, and skin texture.

Encoding and Storage: Each face is encoded into a mathematical model and compared with stored facial data.

Threshold-Based Matching: The system assigns a confidence score to each detected face based on similarity to stored images.

Handling Unrecognized Faces: If a face does not match any database entry, it is marked as an unidentified student or guest.

At this stage, if no match is found, the system records the outcome and halts further attendance marking for that student.

5. *Attendance Recording*

Once student identities are confirmed, the attendance marking process begins:

Open Attendance ExcelSheet: The system accesses the attendance file for the current session.

Mark Present for Recognized Students: Each matched student is marked as 'Present' in the sheet.

Mark Absent for Unrecognized Students: If a student's face is not found in the image, they are marked as 'Absent.'

Handling Special Cases: If astudent is absent but has valid leave approval, the system can differentiate between an excused absence and an unapproved one.

The updated attendance file is then saved with the latest records.

6. *Save and Email Attendance Report*

Once the attendance data is updated, the system generates are port and sends it to the appropriate stakeholders. This step includes:

Generating the Attendance Report: The system formats the attendance data in to a structured report (Excel, PDF, or CSV format).

Adding Time Stamps and Metadata: The report includes session details such as date, time, classroom location, and instructor name.

Email Distribution: The finalized report is sent via email to faculty members, school administrators, and other relevant personnel.

Archiving Data: A back up of the attendance report is stored in a secured database for future reference.

IV. System Description

A. Hardware Components

The attendance management system requires the folloowing hardware and software components. The block diagram in Figure 2 shows the components in the system.

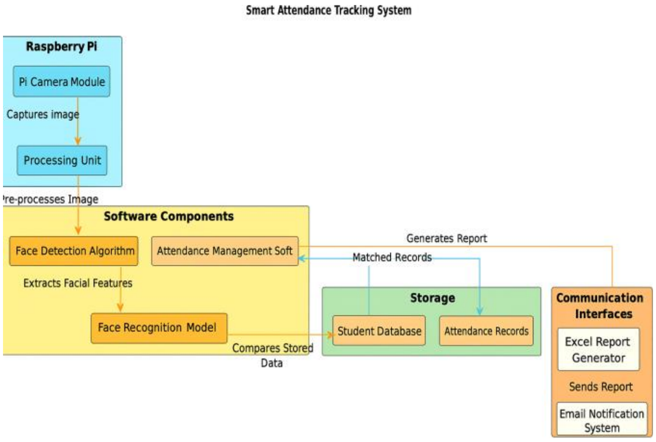


Fig. 2. Block diagram of Smart Attendance System

Raspberry Pi

The Raspberry Pi is the core computing unit that orchestrates the attendance tracking system. It is a compact, low-power, and cost-effective micro-computer used to run the face recognition algorithm and run image processing algorithms.

Pi Camera Module

The Pi Camera Module is responsible for capturing images of students in real-time. It provides high-resolution images to ensure accurate face detection. The camera can be programmed to capture images at specific intervals (e.g., at the beginning of a class session) or when triggered by an event. The module interfaces with the Raspberry Pi through the MIPICSI (Camera Serial Interface), enabling fast image transfer. Image quality parameters such as brightness, contrast, and exposure can be adjusted programmatically for better recognition accuracy.

Processing Unit

The processing unit in the Raspberry Pi pre-processes the captured images before sending them for face detection. It performs tasks like image resizing, noise reduction, and histogram equalization to enhance image clarity. The processing unit may use OpenCV, Dlib, or TensorFlow for face detection and feature extraction. If the system is configured for video-based attendance, the processing unit extracts frames from video streams and processes them individually.

B. Software Components

The software components manage face detection, recognition, and attendance logging. These algorithms use AI and machine learning to analyze the images captured by the Raspberry Pi.

Face Detection Algorithm

This algorithm scans the captured image to identify human faces within it. It utilizes Haar cascades, Histogram of Oriented Gradients (HOG), or deep learning-based Convolutional Neural Networks (CNNs) to detect faces accurately. The algorithm draws bounding boxes around detected faces, filtering out unnecessary details like the background or non-human objects. Face detection models like MTCNN (Multi-task Cascaded Convolutional Networks) or OpenCV's DNN module are commonly used [4, 5, 6, 7, 8, 9].

Face Recognition Model

Once a face is detected, the face recognition model extracts key facial features and compares them with stored student images. The model converts the face into a numerical representation (vector) using deep learning-based feature extraction (e.g., FaceNet, DeepFace, or Dlib). It applies a similarity algorithm (e.g.,Euclidean distance or Cosine similarity) to match the detected face with records in the student database [5, 6, 7]. If a match is found, the student is marked as present; otherwise, the system logs them as unknown or absent.

Attendance Management Software

This module takes the face recognition results and updates the attendance records accordingly. It logs the attendance in an Excelfile, database, or cloud storage. The software can integrate with Institute Management Systems (IMS) to automate attendance tracking and generate reports. It may also use time-stamp based verification to ensure attendance is marked only once per session.

C. Storage

The storage system is critical for managing student records, attendance logs, and historical data. It consists of two major databases.

Student Database

This data base contains student names, rollnumbers, unique facial embeddings (vectors), and other details. It allows quick retrieval of student data when matching detected faces.

Attendance Records

Stores daily, weekly, and monthly attendance logs of students. The records may include additional metadata, suchas date, time, class, and teacher's name. The storage module supports automatic backups to prevent datal oss. The data can be exported in different formats like Excel (.xlsx), CSV,or JSON for easy integration with external applications.

D. Communication Interfaces

Once attendance is processed, the system generates reports and notifies stakeholders (teachers, administrators, parents, etc.).

Excel Report Generator

The system converts the attendance data into structured Excel sheets for easy visualization and analysis. It can generate reports with graphs, charts, and student-wise summaries. The reports may include additional insights, suchas attendance trends, absentee rates, and punctuality analysis.

Email Notification System

This module automatically sends attendance reports to teachers, administrators, or parents via email. The email can contain: Daily or weekly attendance summaries. Alerts for absentee students. The system can be integrated with SMTP (Simple Mail Transfer Protocol).

V. Results And Discussions

This project focuses on reducing the time taken for attendance marking, especially in large entities like universities and corporate offices. Traditional methods such as manual entry, thumbprint scans, and RFID cards are often slow and susceptible to proxy attendance. This system addresses those issues by using facial recognition technology to mark attendance instantly through a simple photo captured via a mobile phone or camera device. It eliminates the need for queueing and significantly minimizes the risk of proxy entries by verifying each face against the stored database. The process is faster, more secure, and more efficient than conventional systems.

As a future enhancement, the integration of YOLO (You Only Look Once) machine learning can be implemented to further accelerate the face detection and recognition process in real-time. The model can also be trained incrementally with each new photo, improving accuracy and performance over time. To handle large datasets and ensure error-free operation, the system can be deployed on a high-configuration server, providing a scalable and human-friendly attendance solution.

In Fig. 3, a classroom setting with a group of students posing for a photo is depicted. The image was utilized as input for the facial attendance recognition system to perform followed by recognition based on pre-trained facial data. This group photo served as a test case to evaluate the system's accuracy face detection and recognition tasks. Each face in the image was automatically detected using the implemented algorithm, in identifying and marking attendance in a real-world classroom scenario.

Fig. 4 shows the screenshot of Python script output after processing a classroom group photo. The system detected 49 faces and recognized several individuals using a pre-trained student database. Key image quality metrics like Laplacian Variance and Average Brightness were also computed. Each recognized person is listed by name along with their student ID. This output confirms successful face detection and recognition for automated attendance marking.

Fig. 5 shows capture of a group of students gathered in an open outdoor environment. The photo was used as input for the facial recognition system to test its performance under natural lighting and non-uniform backgrounds. Despite environmental challenges, the system was able to detect and recognize faces for attendance marking. This scenario demonstrates the system's robustness outside the controlled indoor settings also.

In Fig. 6, the output of the programme shows the names and IDs of students successfully recognized from the outdoor group photo. The system detected and identified multiple faces despite varying lighting and background conditions. This result demonstrates the system's ability to perform accurately in outdoor environments.

Fig. 7 shows the facial recognition system's output applied to a group photo taken at night in an outdoor corridor. Despite low-light conditions, the system successfully detected and recognized five individuals by labeling each face with their name and student ID. This result highlights the system's capability to operate effectively under challenging lighting environments.

In Fig. 8, the Excel sheet displays the attendance record generated from the recognition results of the night-time group photo taken on 28-02-2025. The system matched detected faces with stored data and marked the corresponding students as "Present," confirming accurate performance even in low-light scenarios.

Fig. 3. Group photo of BE-EEE third year Class of 2022-26 Batch (Image 1)

```
>>>
=== RESTART: D:\Studies\SEMESTER 6\INNOVATION PRACTICE\THIRD REVIEW\test 1.py ==
Script started.
Laplacian Variance: 220.10332573269426
Average Brightness: 125.24784733048313
Detected 49 faces in the group photo WhatsApp Image 2025-03-19 at 10.04.06_97cfc8ea.jpg.
TARUNVISHAL I M (22E158) recognized
PON RAMPRASATH P (22E141) recognized
VASUDEV K (22E160) recognized
DHARUN J (22E113) recognized
JAYASUSEENTHAR S (22E125) recognized
SHANKAR K  (22E153) recognized
RAVIRICHARD BALA R (22E146) recognized
SANTHOSH KUMAR K (22E151) recognized
SANUJ S (22E152) recognized
GURUPRASATH M (22E120) recognized
RUDRESH V (22E147) recognized
VARUN PRASAD M S (22E159) recognized
HARI PRASATH (22E120) recognized
VARUN PRASAD M S (22E159) recognized
ANIRUTHAN B (22E106) recognized
```

Fig. 4. Output of the programme from Image 1

Fig. 5. Group photo of third year BE-EEE of 2022-26 Batch in open area (Image 2)

Table I is illustrating the effectiveness of the algorithm by comparing the number of people in the images and identified. It is clearly indicating that, if the face is fully visible in the camera view, the persons are identified easily by the algorithm. So, for identification of people it is necessary to ensure that the face is visible in the image. Again, another snap at different angle or different instant is considered for identifying all people in the classroom.

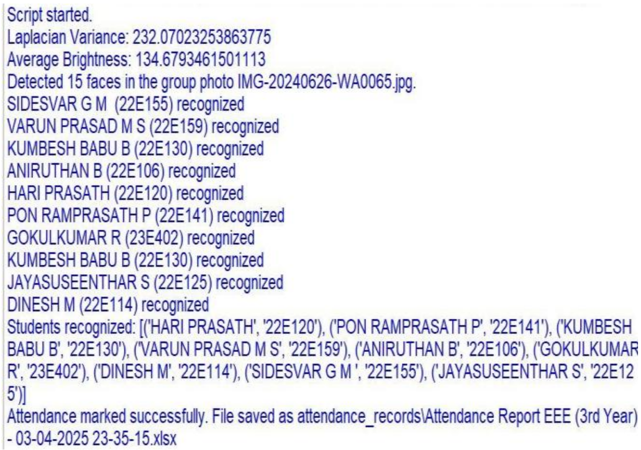
Calculation of metrics for face recognition system

Image 1

This picture is taken in the classroom which contains 50 students' images. The smart attendance system identified the students as per below details.

- Actual students present in image: 50
- Students marked present by the system: 48
- Correctly identified (True Positives, TP): 46
- Missed present students (False Negatives, FN): 4
- False positive recognitions (FP): 2
- True Negatives (TN): Not meaningful for attendance (since absent students are not in photo).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} = \frac{46}{50} = 92\%$$



Script started.
Laplacian Variance: 232.07023253863775
Average Brightness: 134.6793461501113
Detected 15 faces in the group photo IMG-20240626-WA0065.jpg.
SIDESVAR G M (22E155) recognized
VARUN PRASAD M S (22E159) recognized
KUMBESH BABU B (22E130) recognized
ANIRUTHAN B (22E106) recognized
HARI PRASATH (22E120) recognized
PON RAMPRASATH P (22E141) recognized
GOKULKUMAR R (23E402) recognized
KUMBESH BABU B (22E130) recognized
JAYASUSEENTHAR S (22E125) recognized
DINESH M (22E114) recognized
Students recognized: [('HARI PRASATH', '22E120'), ('PON RAMPRASATH P', '22E141'), ('KUMBESH BABU B', '22E130'), ('VARUN PRASAD M S', '22E159'), ('ANIRUTHAN B', '22E106'), ('GOKULKUMAR R', '23E402'), ('DINESH M', '22E114'), ('SIDESVAR G M ', '22E155'), ('JAYASUSEENTHAR S', '22E125')]
Attendance marked successfully. File saved as attendance_records\Attendance Report EEE (3rd Year) - 03-04-2025 23-35-15.xlsx

Fig. 6. Output of the programme from Image 2

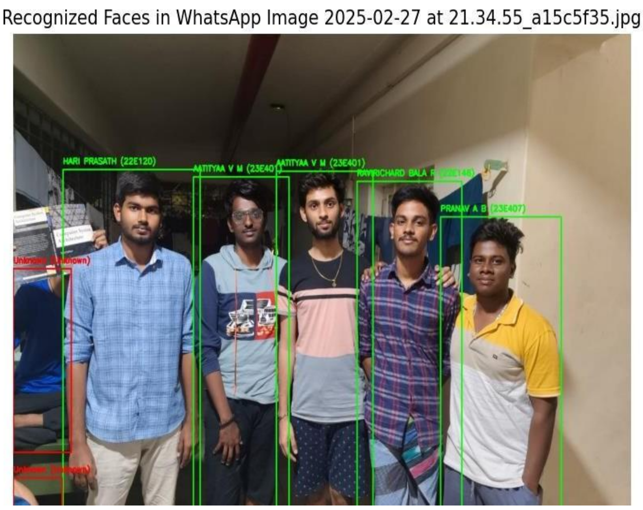
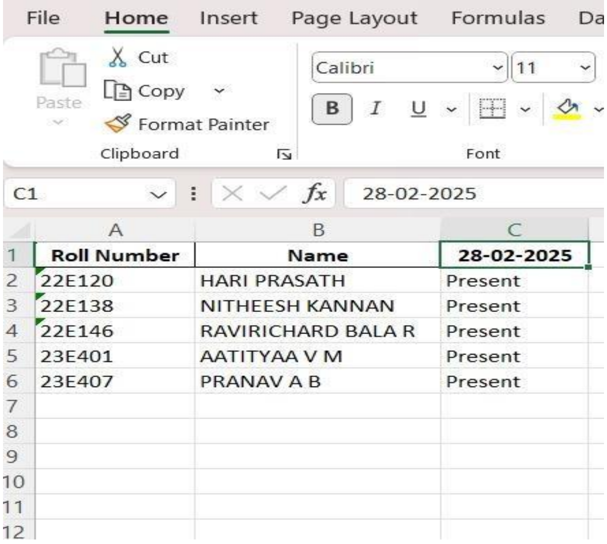


Fig. 7. Photo of third year BE-EEE of 2022-26 Batch (Image 3)



	A	B	C
1	Roll Number	Name	28-02-2025
2	22E120	HARI PRASATH	Present
3	22E138	NITHEESH KANNAN	Present
4	22E146	RAVIRICHARD BALA R	Present
5	23E401	AATITYAA V M	Present
6	23E407	PRANAV A B	Present
7			
8			
9			
10			
11			
12			

Fig. 8. Excel created from Image 3

TABLE I: EFFECTIVENESS OF ALGORITHM

Image No.	No. of Poeple in Image	No. of People identified
1	50	46
2	17	15
3	5	5

$$Precision = \frac{TP}{TP + FP} = \frac{46}{46 + 2} = 95.8\%$$

$$Recall = \frac{TP}{TP + FN} = \frac{46}{46 + 4} = 92\%$$

$$F1\text{-}score = 2 \cdot \frac{Precision \times Recall}{Precision + Recall} = 93.8\%$$

Image 2

This picture is captured in outdoor environment which contains 17 students' images. The smart attendance system able to identify the students as per below details.

- Actual students present in image: 17
- Students marked present by the system: 16
- Correctly identified (True Positives, TP): 15
- Missed present students (False Negatives, FN): 2
- False positive recognitions (FP): 1 *(occasionally, a misidentification or non-student counted due to background noise, hats, or group formation)*
- True Negatives (TN): Not meaningful for group outdoor images.

$$Accuracy = \frac{TP}{TP + FP + FN} = \frac{15}{15 + 1 + 2} = 83.3\%$$

$$Precision = \frac{TP}{TP + FP} = \frac{15}{15 + 1} \approx 93.8\%$$

$$Recall = \frac{TP}{TP + FN} = \frac{15}{15 + 2} \approx 88.2\%$$

$$F1\text{-}score = 2 \cdot \frac{0.938 \times 0.882}{0.938 + 0.882} \approx 90.9\%$$

Image 3

This picture is captured outside hostel room which contains 6 students' images. The smart attendance system identified all the 6 students.

- Actual students present in image: 6
- Students marked present by the system: 6 (all boxed and labeled)
- Correctly identified (True Positives, TP): 6
- Missed present students (False Negatives, FN): 0
- False positive recognitions (FP): 0 (no spurious green boxes)
- True Negatives (TN): Not meaningful for attendance in a group photo (absent people are not in the image).

$$Accuracy = \frac{TP}{TP + FN + FP} = \frac{6}{6 + 0 + 0} = 100\%$$

$$Precision = \frac{TP}{TP + FP} = \frac{6}{6 + 0} = 100\%$$

$$Recall = \frac{TP}{TP + FN} = \frac{6}{6 + 0} = 100\%$$

$$F1\text{-}score = 2 \cdot \frac{Precision \times Recall}{Precision + Recall} = 100\%$$

In Table II, the metrics for all three images chosen for verifying the smart attendance system are listed. From the metrics calculations, it is understood that the proposed system is able identify the faces, if the face is completely visible and has good lighting. The system not able identify the faces which are having shading or partially visible. The system needs to be tested for variety of lighting conditions and with different poses for a period of 6 months. Then the ability of the system to recognize the students assessed correctly and will be implemented in the classrooms.

TABLE II: METRICS OF FACE RECOGNITION SYSTEM

Metric	Image 1	Image 2	Image 3
Accuracy	92%	83.3%	100%
Precision	95.8%	93.8%	100%
Recall	92%	88.2%	100%
F1-score	93.8%	90.9%	100%

VI. CONCLUSIONS

The smart attendance system developed is effectively meets its objective of minimizing time and increasing accuracy in attendance systems by leveraging face recognition technology. The system was tested under various lighting conditions and locations for its performance. The system is detecting the persons based on the database stored and updates the presence in the excel file and the report is sent to the concern in-charges through email. By enabling attendance through simple camera input and proposing future integration with YOLO for real-time detection and adaptive learning, the system offers a scientifically effective, efficient, and humanized solution to traditional attendance challenges.

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